

The effects of campaign spending on electoral outcomes: A data envelopment analysis*

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Abstract. Data envelopment analysis is used to analyze the efficiency of candidate campaign spending. Analysis of the measured efficiency scores of the incumbents and challengers shows that there are significant systematic effects of both district and candidate characteristics. The incumbent's residuals from the efficiency score regressions indicate those candidates who do well because of the nature of their district and those who do well because of their own abilities. Incumbents are ranked according to their observed efficiency, their unobserved efficiency and their vote share. Correlations among the rankings are significant and of the expected signs. More efficient candidates receive higher vote shares.

1. Introduction

The early literature on the efficacy of campaign spending developed a production theory relationship between candidate spending and vote share. Subsequent research has at least implicitly maintained the production model, though without further theoretical analysis of the relationship. If one accepts the production model, then asking it about the efficiency of the producers is natural. For example, are producers getting the most "output" from their inputs, or are some being wasted? This is not the same thing as asking what the marginal productivity of spending is, however, because the observed marginal productivity need not be the efficient marginal productivity. The literature to date has focused its attention exclusively on measuring the observed productivity of spending. The literature has not addressed the question of what level of productivity is possible from the given resources.

This paper uses a technique known as data envelopment analysis, DEA, to measure the actual practice of candidates relative to the best observed practice. For example, if one candidate generates a 54% vote share spending

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\$250000, and the best any candidate does with that much spending is a 60% share, then the first candidate is rated with an efficiency score of 1.111 and the second a score of 1. These figures indicate by what factor a candidate's output must be increased, given her spending level, for her to replicate the best observed practice, that is, be on the efficient frontier. Of course, few candidates spend precisely the same amount as any others, so a method must be found of determining which candidates with whom to compare any given individual. The DEA technique is such a method that uses linear programming to obtain a piecewise linear production frontier; it is a deterministic approach in contrast to the stochastic regression method. By its nature, DEA shows that some producers are totally efficient; no other producer or set of producers generates a greater output with the same or a lower amount of resources. Unlike regression, no observed point is above the estimated production frontier.

Of course, in our example above, attributing all of the vote share differential to inefficiency may be an exaggeration because other influences on the elections, which are beyond the control of the candidates, may differ. For example, a Congressperson does not control the median income level or the urbanization within her district, the racial composition of her constituents, their educational attainment, or their "ideology." Candidates have no latitude to vary their race or age, nor can they change their educational background or gender during the campaign. Moreover, no candidate chooses the spending by her opponent. Each of the district or personal characteristics and opponent spending may, however, systematically influence the efficiency of one's own campaign expenditures. A second stage analysis of the efficiency scores both reveals the systematic impact of factors outside the control of the candidate and allows calculation of a pure efficiency score – one purged of the uncontrollable influences. Examination of these pure efficiency scores reveals which candidates do well because they use well the resources they have and which do well despite themselves.

The finding that district and individual characteristics influence the ability to produce votes from spending should not surprise anyone. Nonetheless, very little of the research on the link between campaign spending and electoral outcomes considers such factors except as they bear on the strength of the candidate's party within the district or the quality of the challenger. The usual specification in the literature treats district party strength and challenger quality as though their only effect is to shift the intercept in the regression equation, ignoring the possibility that these variables influence the slope of the relationship between campaign spending and vote share.¹ The DEA results suggest, however, that a more careful consideration of the influence of district and candidate attributes is necessary if one is to understand the spending vote share relationship. In fact, these results indicate that such

characteristics might influence the parameters of the vote share production function; that is, variation in individual and district characteristics might cause variation in the effect of additional spending on vote share.

The second-stage analysis also is useful for another reason bearing on the debate in the literature about how to measure challenger quality. Jacobson (1978, 1985) uses a simple dummy variable showing whether the challenger has or has not previously held public office. Green and Krasno (1988, 1990) take Jacobson to task for this mismeasurement of challenger quality. They construct an index of challenger quality in which various challenger attributes produce quality points. Their challenger quality variable appears to have a strong impact on estimates of the effects of campaign spending. Unfortunately, their index introduces unknown biases because of its arbitrary and ad hoc nature (Levitt, 1994). The pure efficiency constructed from the second stage analysis of the DEA efficiency scores does not suffer from this defect. In particular, an increase in this variable of 10% indicates precisely a 10% decline in the efficiency of the challenger's campaign.

Interestingly, the literature does not address the question of incumbent quality. Green and Krasno (1990) note this about their own work but do nothing about it.² Jacobson (1985) suggests, however, that once a challenger has attained a threshold level of recognition and regard among the constituency, largely through her own spending, the outcome of the race depends upon the content of the campaigns. Therefore, the ability of a campaign effectively to use its resources becomes crucial. In a tight race this must be true for both the challenger and the incumbent. Therefore, omission of incumbent quality might have substantial impacts on vote share spending relations. The pure efficiency of the incumbent captures the campaign effectiveness better than other variables frequently used in vote share equations. For example, regressions often include variables such as terms in office, or vote share in the last election, which show incumbent abilities as a campaigner, but also suggest other desirable attributes in a candidate and even district ideology and partisan tides. The pure efficiency of a candidate contains none of these influences.

A final reason for interest in the second stage analysis is what information it conveys about specific candidates. The pure efficiency score suggests which candidates do well because of their district, and which do well because they use resources wisely. If the national party is concerned about allocating scarce funds among candidates, and wishes to maximize the number of seats in the legislature, then it should contribute to those of its candidates in tight races, or holding marginal seats, most capable of putting the funds to good use. Contrast this with standard vote share models that cannot provide this information because the effects of additional spending are constant across candidates, ex-

cept in as far as their spending levels differ and diminishing marginal returns are built into the model. Moreover, the unexplained portion of the vote share, i.e., the regression error, is, if the model is correctly specified, uncorrelated with the effects of spending, or any other explanatory variable. It is simply a random shock to the incumbent's vote share that may be the effect of voting booth troubles, rain, high or low interest in other issues on the ballot, or any number of other similar sources of variation in turnout and participation in the election specific to the district. Consequently, this regression error from the standard approach does not carry a natural interpretation with practical significance.

The paper is organized as follows. Section 2 explains the DEA technique and the theory behind the analysis of the efficiency scores. Section 3 discusses the campaign spending vote share data, the results of the DEA model, and uses the computed efficiency scores from the DEA analysis as the dependent variable in an econometric investigation of the sources of (in)efficiency in producing electoral outcomes. Selecting candidates revealed to be either extremely ineffective or extremely effective campaigners, the fourth section compares the evidence from this study with the discussion of these candidates in the *Almanac of American Politics 1986*. The fifth section concludes.

2. Introduction to DEA

2.1. DEA theory

The basic approach to DEA is to use linear programming to fit a surface to the upper most data points.³ The programming problem determines which of the data points are undominated by any others. More formally, the analysis finds the supporting hyperplanes of the data. Suppose that the data points are defined as (y_i, x_i) where y is an m -vector of outputs and x is an n -vector of inputs of observation i , of which there are I^* .⁴ The space of the model is, therefore, R^{m+n} – the $m+n$ dimensional Euclidean space. A hyperplane in this space is given by $\mu y_i - \delta x_i + \omega = 0$ where μ and δ are appropriately dimensioned vectors and ω is a (scalar) constant. The production surface is portions of a set of hyperplanes each of which satisfies the following:

$$\begin{aligned} \mu y_i - \delta x_i + \omega &\leq 0 \text{ for all } i \\ \mu y_k - \delta x_k + \omega &= 0 \text{ for some } k. \end{aligned}$$

In words, all the data points lie below or on the hyperplane defined by the parameters μ , δ and ω and at least one data point lies on the hyperplane. Therefore, one method of solving the problem of finding the surface is to maximize the expression $\mu y_j - \delta x_j = \omega$ by choosing μ , δ and ω subject to the constraints $\mu y_i - \delta x_i + \omega \leq 0$ for all i , that the constraint is satisfied as an

equality for at least one observation, and that $\mu_m \geq 1$ for all m and $\delta_n \geq 1$ for all n . Note that the optimization problem is solved for each individual observation; hence, I^* linear programming problems are solved.

Alternatively, one might solve the dual problem, which may be more intuitive.⁵ Here, the objective is to minimize the slack in the resources and the output. Write slack in an output as s and slack in an input as e . The objective function is now to minimize $-(\sum s_m + \sum e_n)$ by choosing the s_m , e_n , and a vector of λ 's subject to several constraints. Along with nonnegativity of the s , e , and λ 's, these constraints are:

$$\begin{aligned} \sum y_{mk} \lambda_k - s_m &= y_{mi} \text{ for each } m, \\ -\sum x_{nk} \lambda_k - e_n &= -x_{ni} \text{ for each } n, \text{ and} \\ \sum \lambda_k &= 1 \end{aligned}$$

where the summations are over the number of DMUs.⁶ Because the λ_k are nonnegative and sum to one, each is also less than or equal to one. The λ_k 's are, therefore, weights, so that the output (input) of observation i is compared with a weighted average of the output (input) of the other observations. For an efficient observation j , $s_m = 0$, $e_n = 0$ for all m and n ; efficient units do not experience slack in any outputs or any inputs. For DMU's at the corners of the production surface, through which more than one supporting hyperplane passes $\lambda_j = 1$, and $\lambda_k = 0$, $k \neq j$; such units are productively efficient by default. However, a unit that is efficient may be so without $\lambda_j = 1$, and $\lambda_k = 0$, $k \neq j$. The outputs and inputs of these DMU's are linear combinations of those of a subset of the units that are efficient. For the linear program of a given DMU, those DMU's with nonzero λ are called the reference set, it is these to which the given DMU is compared.

Consider Diagram 1 in which there are one input and one output. The dashed lines are the hyperplanes found using the DEA analysis. The production surface is the piecewise linear frontier given by OABC extending horizontally from point C.⁷ A, B, and C are each input-output combinations associated with the particular DMUs A, B and C. Diagram 2 shows the surface more clearly, and includes an inefficient DMU D. Points E and F on the frontier are linear combinations of the input-output combinations found at DMU A and B. These points, E and F, show respectively the output an efficient firm would obtain using the input of DMU D and the input an efficient firm would require to produce the output of DMU D. The ratio of the height to point E to that of point D is the efficiency score of DMU D. For simplicity, these efficiency scores will be denoted by TE_i . Note that for an efficient unit the efficiency score is unity, but for inefficient DMUs it is greater than unity.⁸ An efficiency score of 1 shows that the DMU could not expand output without

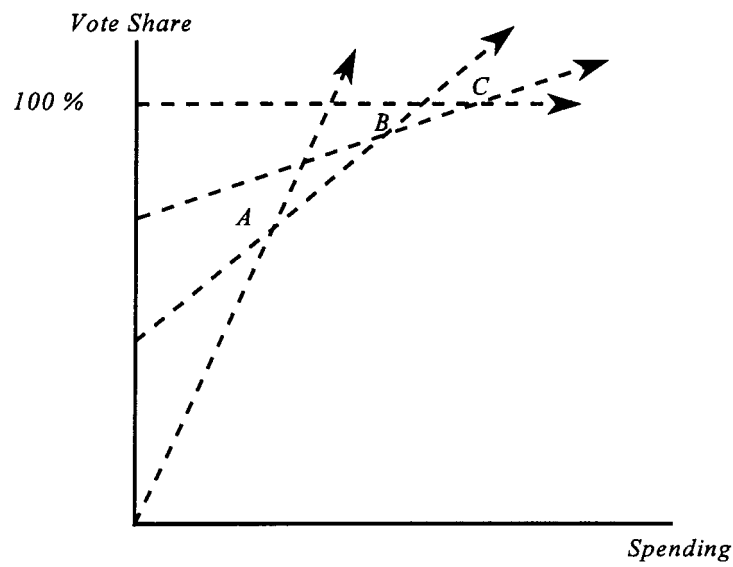


Diagram 1.

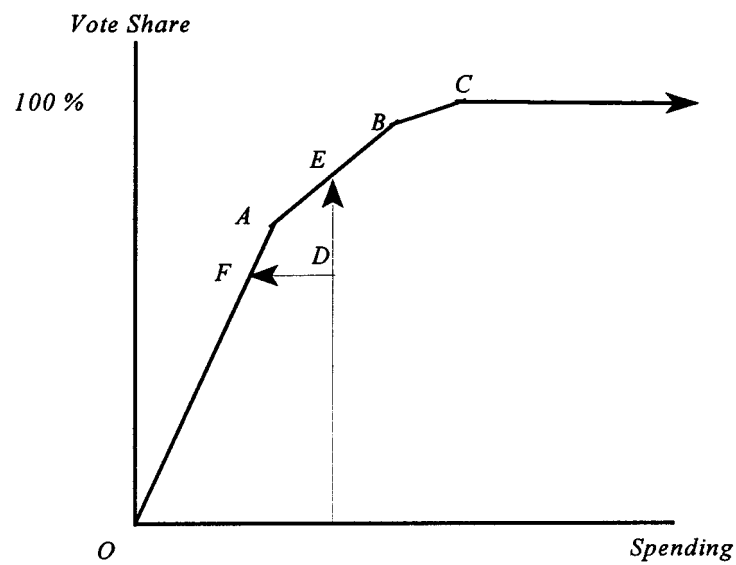


Diagram 2.

additional input; scores greater than 1 show that output could be increased proportionally by that amount without increasing input usage. For example, if $TE_i = 1.85$, then for that DMU to be efficient it must increase its output 1.85 times without increasing its inputs.

These simple diagrams are especially appropriate in the current context because the model here involves only one input, candidate campaign spending, and one output, vote share. This assumption, however, does not mean that only own spending influences the election results. Instead, additional factors upon which a candidate's vote share may depend are introduced in a second stage regression analysis of the estimated efficiency scores. The nature of this analysis is described in greater detail below.

A debate about the appropriate form of DEA analysis exists in the literature. On the one hand, some argue that all the determining factors belong in the DEA. Techniques exist for including nondiscretionary and categorical variables in the model (Banker and Morey, 1986). On the other hand, in the current situation it is not clear what the expected impact of the additional variables would be. For example, the direction of the effect of a highly urban district on ease of vote share production for an incumbent is not clear. Moreover, it may vary with the party of the incumbent. Consequently, the manner in which these variables enter the production function is not clear a priori. Including them in the DEA is, therefore, problematic. As a result, variables such as district and personal characteristics are used in the regression analysis of the technical efficiency scores.

2.2. Analysis of DEA results

A set of variables not under the control of the DMU explain the variation in the TE. For the case of campaign spending, let z_i be a vector of district and individual specific variables that are not choice variables of the candidate during the current campaign, β is a vector of coefficients.

$$TE_i = z_i\beta + \mu_i \quad (1)$$

The $z_i\beta$ portion of the regression model is that part of the efficiency score produced by observed factors outside the control of the candidate.^{9,10} So, estimates of the residual μ show that portion of the efficiency score that is idiosyncratic to the candidate, the election, or other omitted variables, purely random effects, or measurement error. Note that estimates of μ can be positive or negative. A DMU whose estimated μ is negative is one that looks inefficient because of factors outside its control; that is, for which the systematic part of the efficiency score overstates the true inefficiency of the unit. The more negative the estimated residual, the more truly efficient the DMU. If the estimated residual is positive, the systematic component of the efficiency score understates the true inefficiency of the candidate. The larger the estimated residual, the greater the true level of inefficiency of the campaign.

Consistent with the omitted variables explanation for the source of the regression error, the residual from the estimation of TE may be interpreted as a measure of the candidate's ability as a campaigner, or as her effectiveness as a legislator. Under this interpretation, the residual shows how capable the candidate is at turning campaign expenditures into vote share, independent of all the district and individual qualities that influence the election outcome. In the results section below, this interpretation of the residual is compared against, and found consistent with, the assessments of several candidates taken from the *Almanac of American Politics 1986*. These candidates are chosen from among both the worst and the best according to the efficiency scores.

There are (at least) two additional intuitive explanations of the residual, each of which is consistent with an omitted variable as the source of the regression error. First, it is clear from the discussion above that the DEA technique accommodates multiple outputs and multiple inputs. Its great advantage over stochastic estimation methods is precisely its ability to analyze a production model with joint products. In the vote share campaign expenditures setting of this paper, several natural, but unmeasurable, joint products with vote share are imaginable. For example, a candidate might be advancing a policy agenda such as tax reform or universal health care. Alternatively, the incumbent might be laying the groundwork for an attempt at higher statewide office or a run for the presidency. Such possibilities might explain large expenditures by incumbents without accompanying increases in vote share. In any of these cases, the technical efficiency TE of this incumbent will be affected adversely, suggesting a less efficient campaign than is actually the case. Therefore, the estimated residual from Equation (1) may represent, in part, the influence of these unaccounted for joint products on the efficiency score of a candidate.

The second additional intuitive explanation of the residual in the efficiency score equation is what John Lott (1986, 1991) has termed political brand name capital. He suggests that candidates build brand name through expenditures and reputation. Lott (1986) argues that the repeated findings in the literature of a negative correlation between incumbent expenditures and incumbent vote share can be explained by the omission of this brand name capital from vote share campaign expenditure equations. In the DEA analysis, this brand name capital is an omitted input, whose effects will show up in the residual of the efficiency score Equation (1).

From a political standpoint, knowledge of these residuals is valuable. Consider a party with limited resources deciding which among various candidates it should support with more funds. Candidates whose seats are "marginal" in the sense that their previous vote share was small are those whose prospects can be helped most by an infusion of money. The party will want to allocate its funds between the marginal candidates to maximize its total seats. Hence,

the party would like to know which of its candidates would use the funds most effectively. The estimated residuals may provide this information.

3. Data and DEA results

The data set analyzed includes all those incumbent members of the House of Representatives seeking reelection in 1984 who faced a challenger whose spending was nonzero. Therefore, there are no open seat elections, no unchallenged incumbents, and no incumbents whose opponent spent nothing on her campaign. In 1984, 288 House races satisfied these criteria. Additionally, the data set includes socio-demographic and economic characteristics of the legislative districts and personal characteristics of the incumbents.¹¹ Table 1 defines the relevant variables and provides their descriptive statistics.

Efficiency scores are computed for each of 288 incumbents and 288 challengers using an output oriented variable returns model.¹² The scores range from 1, complete technical efficiency, to 5.163 for challengers and to 2.068 for incumbents. The mean incumbent efficiency score is 1.434 with a standard deviation of .23; for challengers the values are 2.118 and .519, respectively. Table 2 presents these descriptive statistics on the efficiency scores. Of the 576 observations, five are efficient; three of the five are challengers. The efficient incumbents are William Gray and Robert Walker, both of Pennsylvania. The efficient challengers spent very little money, less than \$1000, and opposed powerful Democrats (Bonoir and Dingell of Michigan, and Pease of Ohio). None of the three efficient challengers garnered more than 40% of the vote.¹³

The efficiency scores in themselves are far less interesting than an examination of their sources. Recall from Equation (1) above that a variety of factors outside the control of the candidate are used to predict the efficiency score. Table 3 presents results of OLS regressions using district and personal characteristics variables, and opponent expenditures to predict candidate efficiency. The results allow for tests of several hypotheses. First, one can test for the significance of district characteristics given the presence of personal characteristics of the incumbent, opponent spending, and the incumbent's lagged vote share. Second, one can test for the significance of personal characteristics given the use of district characteristics, opponent spending and lagged vote share.¹⁴ Third, one can test for the significance of both types of variables given opponent spending and the lagged vote share of the incumbent. Predicting challenger efficiency using incumbent spending, incumbent electoral history and district characteristics is also possible. A fourth hypothesis is, therefore, that district variables influence challenger efficiency even after controlling for incumbent spending.

With respect to these four hypotheses the answers are resounding in their acceptance of the district and personal characteristics variables as holding

Table 1. Variable definitions and descriptive statistics

Variable	Mean	SD
DEMPRES: Vote share of Carter within the district in 1980	39.98	9.35
PCTWHITE: Whites as share of district population	85.79	14.59
MEDAGE: Median age of district population	30.33	2.93
HSEVAL: Median house value in the district	53.41	23.57
MEDINC: Median income in the district	20.60	3.92
PCTBLUE: Blue collar workers as share of district population	31.03	7.14
PCTURB: Urban population as share of total population	74.50	26.98
PCTOWN: Homeowners as share of total population	65.91	11.75
PCTCGRAD: College graduates as share of total population	16.66	6.50
SOUTH: 1 if state of Confederacy, 0 otherwise	.184	.388
WEST: 1 if a western state, 0 otherwise	.254	.436
AGE: Age of the incumbent	47.90	10.26
GENDER: 1 if the incumbent is male, 0 otherwise	.938	.242
RACE: 1 if the incumbent is non-white, 0 otherwise	.049	.215
MARRIED: 1 if the incumbent is married, 0 otherwise	.826	.379
LAWYER: 1 if the incumbent is a lawyer, 0 otherwise	.483	.501
ADA: The incumbent's ADA rating	50.54	35.32
TERMS: The number of terms served by the incumbent	3.06	3.16
LAGVOTE: The incumbent's vote share in the last election	63.92	11.23
DEM: 1 if the incumbent is a Democrat, 0 otherwise	.576	.495
DEM*DEMPRES: Interaction between incumbent's party and the district's support for Jimmy Carter in 1980	25.10	22.59
RACE*PCTWHITE: Interaction between the incumbent's race and the share of the district that is white	2.72	13.52
INC\$: Incumbent spending in tens of thousands of dollars	30.40	18.48
CHAL\$: Challenger spending in tens of thousands of dollars	13.65	18.85

Table 2. Efficiency scores

	Mean	SD	Minimum	Maximum
Incumbent	1.4343	.2298	1.0	2.068
Incumbent pure efficiency	0	.1207	-.3781	.5704
Challenger	2.118	.5189	1.0	5.163
Challenger pure efficiency	0	.4354	-1.026	2.082

Table 3. Efficiency score regressions

Variable	Incumbent regressions				Challenger regression	
	Model 1	Model 2	Model 3	Model 4	Model 1	Model 2
Constant	1.56 (.05)*	1.59 (.18)*	1.58 (.08)*	1.45 (.20)*	1.25 (.24)*	2.11 (.67)*
DEMPRES		-.19 (.14)		.27 (.19)		.14 (.06)*
PCTWHITE		.16 (.1)		.03 (.11)		-.11 (.04)*
MEDAGE		-.40 (.35)		-.37 (.34)		.61 (1.2)
HSEVAL		.06 (.08)		.04 (.08)		.19 (.25)
MEDINC		.19 (.48)		.16 (.47)		-.38 (.16)*
PCTBLUE		.04 (.22)		.14 (.22)		-.18 (.08)*
PCTURB		.02 (.04)		.02 (.04)		-.07 (.13)
PCTOWN		-.05 (.12)		-.04 (.12)		.11 (.04)*
PCTCGRAD		-.32 (.29)		-.12 (.29)		.67 (1.0)
SOUTH		.08 (.03)*		.07 (.03)*		-.14 (.09)
WEST		-.03 (.03)		-.03 (.03)		.05 (.09)
LAGVOTE	-.50 (.08)*	-.48 (.08)*	-.52 (.08)*	-.44 (.08)*	.11 (.03)*	.72 (.28)*
CHAL\$	.13 (.01)*	.12 (.01)*	.12 (.01)*	.11 (.01)*	.33 (.46)	.37 (.44)
CHAL\$ ²	-.68 (.08)*	-.61 (.09)*	-.66 (.08)*	-.61 (.09)*	-.27 (.46)	-.25 (.44)
DEM*				-.83 (.23)*		-.21 (.76)*
DEMPRES						
RACE*				.07 (.16)		
PCTWHITE						
DEM	.09 (.02)*	.11 (.02)*	.13 (.04)*	.43 (.10)*	.30 (.06)*	1.15 (.30)*
AGE			.03 (.09)	.07 (.09)		
GENDER			-.84 (3.3)	-.02 (.03)		
RACE			-.08 (.04)*	-.08 (.11)		
MARRIED			.31 (2.1)	.32 (2.1)		
LAWYER			-.03 (.02)**	-.04 (.02)*		
ADA			-.05 (.05)	.91 (5.8)		
TERMS			.43 (.31)	.33 (.31)		
ADJ. R ²	.67	.68	.67	.70	.09	.25

^a Indicates that this coefficient and standard error have been multiplied by 100.

^b Indicates this spending is that of the incumbent.

* Coefficient significant at the 5% level. **Coefficient significant at the 10% level.

explanatory power for the efficiency of the incumbent, and of the district variables having explanatory power for the efficiency of the challenger. In fact, for each of the four null hypotheses of no joint significance, rejection is at the 1% level. It is clear, therefore, that personal characteristics of the incumbent and district level characteristics influence the efficiency with which both incumbents and challengers produce votes with their campaign spending.¹⁵

Examining the regression results in Table 3 one might feel uneasy about the lack of individually significant variables, especially in the incumbent regressions. One explanation for these results might be high multicollinearity. Of course, the joint significance tests are the appropriate method of determining whether the variables belong; nonetheless, finding more variables individually significant might show which variables in a group are more important. Another possible explanation, however, is that the incumbent and challenger efficiency scores are related. For example, it might be surmised that the error term in the incumbent equation must be correlated with the error term in the challenger equation. In this context, ordinary least squares estimates are unbiased but inefficient; that is, they do not have the least variance. A seemingly unrelated regression model accounts for the error correlation, thereby generating minimum variance estimates.¹⁶ Table 4 presents these results.

The SUR model does, in fact, bring several variables into individual significance that were not so under OLS. In the incumbent efficiency score equation, DEMPRES, GENDER, and TERMS all become significant. However, LAWYER becomes insignificant in the incumbent equation. The SUR model raises both spending variables in the challenger equation and SOUTH to significance. So, as suggested, accounting for the potential correlation in the errors in the incumbent and challenger efficiency score regression leads to the result that several variables are individually significant. Moreover, one cannot reject the first, third and fourth hypotheses above within the SUR model.

The signs of specific coefficients are also informative. For example, the coefficients on DEMPRES and DEM*DEMPRES have opposite signs in both the incumbent and challenger equations. The positive coefficient on DEMPRES shows that the stronger the support for Jimmy Carter within the district, all other things equal, the higher the raw technical efficiency score of the Republican candidate, that is, the less efficient the campaign appears. The negative sign on DEM*DEMPRES shows that the Democratic candidate from a Democratic stronghold tends to appear to be a more effective campaigner than she in fact is. These variables, therefore, capture the impact of a candidate's party's strength in the district on the effectiveness of the campaign.

Table 4. Seemingly unrelated regression results

Variable	Incumbent	Challenger
Constant	1.38 (.19)*	1.73 (.64)*
DEMPRES	.31 (.18)**	.14 (.06)*
PCTWHITE	.14 (.11)	-.11 (.03)*
MEDAGE	-.38 (.33)	.77 (1.2)
HSEVAL	.08 (.07)	.19 (.24)
MEDINC	.11 (.45)	-.36 (.16)*
PCTBLUE	.15 (.21)	-.15 (.07)*
PCTURB	.02 (.04)	-.01 (.13)
PCTOWN	-.08 (.12)	.97 (.42)*
PCTCGRAD	-.16 (.28)	.81 (1.0)
SOUTH	.08 (.03)*	-.17 (.09)**
WEST	-.03 (.03)	.06 (.09)
LAGVOTE	-.40 (.08)*	.87 (.27)*
CHAL\$	.13 (.01)*	.13 (.38)*
CHAL\$ ²	-.75 (.08)*	-.12 (.04)*
DEM*DEMPRES	-.91 (.22)*	-.21 (.07)*
RACE*PCTWHITE	-.17 (.14)	
DEM	.47 (.09)*	1.17 (.29)*
AGE	.01 (.08)	
GENDER	-.06 (.03)*	
RACE	.14 (.10)	
MARRIED	.86 (1.8)	
LAWYER	-.22 (.14)	
ADA	-.02 (.05)	
TERMS	.47 (.26)**	

*Coefficient significant at the 5% level. **Coefficient significant at the 10% level.

Consider also the race variables. Challengers are seen to be significantly more efficient, have lower technical efficiency scores, the greater is the proportion of the district population that is white. For the incumbents, neither PCTWHITE, RACE*PCTWHITE nor RACE is individually significant. Under the SUR model, however, they are jointly significant. Moreover, the signs of the coefficients make intuitive sense. As PCTWHITE rises, a nonwhite incumbent becomes less efficient; a white incumbent, however, appears to be a more efficient campaigner in a district that has a higher proportion of whites. Interestingly, holding constant the racial composition of the district,

white candidates appear to be less effective campaigners than nonwhites; that is, RACE has a positive sign. Why this should be so is unclear, though it may be that white candidates are more likely to be laying the foundation for seeking higher office or pushing a policy agenda. Another interpretation is that white candidates face lower obstacles than nonwhites in becoming candidates, allowing weaker white campaigners to succeed while nonwhites of the same ability would fail. Finally, nonpolitical opportunities for whites may attract the best whites, while politics may be a (relatively) more attractive career for nonwhites.

Summarizing the effect of other district characteristics on the efficiency score of the incumbent, incumbents are clearly more efficient in districts where the population is older, where homeownership is greater, where educational attainment is greater and in the west. Incumbents have a harder time producing vote share in wealthy urban districts, where HSEVAL, MEDINC, and PCTURB are large, and where blue collar employment PCTBLUE is high.

Among the personal characteristics of the incumbent, GENDER, and TERMS are individually significant, as mentioned above. GENDER has a negative sign, indicating that male incumbents appear more effective in producing votes from spending. Candidates also are less efficient the more terms in office they have served. This may reflect the likelihood that more senior representatives spend more time advancing a policy agenda or are involved in preparing for a run at higher office. It may also be that more senior representatives do not take advantage of modern campaign techniques or are not as physically, mentally, and emotionally capable of the rigors of campaigning. This latter possibility is consistent with the positive coefficient on AGE.

The LAWYER variable has negative sign, showing that lawyers are better campaigners than nonlawyers. Why this should be is not clear, though it may be evidence of self-selection of those interested in a career in politics into the law. It may also be that law school teaches the type of skills that make one an effective campaigner. MARRIED and ADA are clearly not significant.

Recall that the residuals from the efficiency score regressions may be interpreted as the “pure” efficiency of the DMU. In other words, the residual is the contribution to the efficiency score made by the individual candidate’s management of her campaign after accounting for the factors outside her control. Table 2 reports the means, standard deviations, minima and maxima of these pure efficiencies for the incumbents and the challengers. As one would expect, incumbents are, generally, more effective campaigners than challengers. This is evident by both the much lower maximum pure efficiency score for incumbents and by the smaller standard error for incumbents.¹⁷

Splitting the sample into incumbents that lost and those that won emphasizes the difference between winners and losers. Incumbents that lost had much larger values of TE and of the estimated residual than did winning incumbents. For example, of the 17 incumbents from the sample who had vote share less than 50%, mean TE is 1.9293 with a standard deviation of .0777; mean residual TE is .14016 with standard deviation .1773. Moreover, only 3 of these 17 had a negative residual from the OLS estimation of the measured technical efficiency score, indicating these individuals are effective campaigners in hostile districts. The challengers of these 17 incumbents have mean TE of 1.7605 (.0773) and residual of $-.1228$ (.2290). For the 271 incumbents whose share of the vote is 50% or more, the means are 1.4032 and $-.0088$, respectively and the standard deviations are .1984 and .1109. For the 141 of these who had negative residuals, the mean residual is $-.0913$ and the standard deviation is .0724. The mean TE of the challengers of the 271 incumbents whose vote share is at least 50% is 2.1404 (.5266) and the mean residual TE is .0077 (.4443). Additionally, 129 of these challengers have a residual less than zero. Interestingly, in about 36% of the races both candidates' residuals are of the same sign; in 17.4% of the cases both are negative, in 18.4% both are positive. One could interpret this as indicating that about 36% of the time both candidates are of the same general ability.

An oft made argument is that high quality incumbents do not face high quality challengers. To the extent that high quality incumbents run unopposed this is irrefutable. The evidence here confirms the hypothesis, however, even in contested elections; the estimated correlation between residuals is $-.026$. Moreover, formally testing if this correlation is zero, the null hypothesis of independence is easily rejected at any conventional significance level.¹⁸

These results, and the others discussed so far, have shown that the DEA approach has some value in the study of vote share campaign spending relations. One might doubt whether the estimated residuals are a true or accurate measure of candidate quality however. In the next section, the analysis compares the evidence from the approach above with descriptions of the electoral circumstances for several candidates.

4. Candidate comparisons

A natural question to ask about the technical efficiency scores from the DEA and the residual components of those scores is how well they stack up against perceptions by analysts of Congressional elections. To address this question, this section provides evidence on which incumbents seeking contested reelection in 1984 appear to be most, and which appear to be least, efficient at using campaign funds. In addition, a spotlight is aimed at several specific legislators revealing that the residuals do a good job of showing candidate effectiveness.

Tables 5 and 6 show the least and the most efficient incumbents and their rank under vote share, technical efficiency score and estimated residual efficiency score.¹⁹ The lower the ranking, the smaller the value of the variable on which the ranking is based. For example, Robert Walker R-PA, is the most efficient incumbent based on the efficiency score from the DEA. Both his TE and his rank are 1. However, seven incumbents have estimated residuals smaller than Walker; hence his rank in the RESID column is 8. Two hundred and sixty-four of the 288 incumbents in the sample have lower vote share than does Walker.

The TE and RESID columns illustrate the point that some candidates do well because, and others in spite, of their districts. For example, William Gray D-PA has a raw efficiency score of 1, suggesting no one does better in generating vote share than he without also spending more money. Yet 182 of the 288 incumbents have smaller RESID; under the interpretation offered here, Gray was slightly worse than average in effectively using campaign spending to generate vote share. Robert Walker R-PA, on the other hand, also has a raw efficiency score of one, but RESID of $-.234$, the eighth smallest. Walker appears to be a very effective campaigner.

Looking at Table 6 reveals who were ineffectual campaigners because, apparently, of their districts and who were ineffectual because of their own skills. For example, Dan Crane R-IL and Don Albosta D-MI are each among the 20 least effective campaigners according to their raw efficiency scores. However, Crane gets there by way of his own lack of campaign skills evidenced by his RESID of $.357$, which results in a net (of RESID) raw score of only 1.5791 less than one standard deviation above the mean. On the other hand, Albosta is in this group despite his strong campaign skills revealed by his RESID of $-.082$, implying a net (of RESID) raw score of 1.939 . To emphasize that this innate ability to use campaign funds effectively matters for real elections, note that while Crane lost by 5% of the vote to an opponent he outspent by almost \$82000, Albosta lost by 1% of the vote to a challenger who outspent him by \$328067. Note also that Crane was a Republican and could be assumed to have gained from the simultaneous reelection of Ronald Reagan to the presidency, while Albosta was fighting that "partisan tide."

On the other hand, one might contend that Crane's residual is really measuring the influence of his involvement in a sex scandal with a female House page. During the campaign, the *Almanac of American Politics 1986* notes, he made stops throughout the district, family in tow, in which he apologized for his behavior. The same article also notes that Crane's legislative record was one of the weakest in the House. In contrast, Gerry Studds, D-MA, was implicated in a scandal in which he had sex with a male House page. Moreover, he was up for reelection in what the *Almanac* described as a Republican dis-

Table 5. Most efficient incumbents as measured by TE and TE residuals

Name and state	Value		Rank		Vote
	TE	RESID	TE	RESID	
Walker, R. PA*	1	-.2340	1	8	265
Gray, W. PA	1	.0344	1	183	288
Owens, R. NY	1.0111	-.0660	3	76	287
Goodling, B. PA*	1.0323	-.2185	4	9	251
Jacobs, A. IN*	1.0328	-.3781	5	1	73
Applegate, D. OH*	1.0330	-.1813	6	18	251
Savage, G. IL	1.0393	-.0267	7	117	280
Young, C.W. FL	1.0400	-.1117	8	49	276
Nowak, H. NY	1.0423	-.1029	9	55	265
Murphy, A. PA*	1.0466	-.1847	10	15	269
Broomfield, W. MI*	1.0521	-.2138	11	11	275
Smith, V. NE	1.0662	-.0510	12	90	283
Towns, E. NY	1.0705	-.0110	13	131	286
Miller, C. OH*	1.0806	-.2436	14	6	236
Cooper, J. TN*	1.0827	-.2600	15	4	251
Lott, T. MS	1.0833	-.1724	16	21	285
Kindness, T. OH*	1.0916	-.1744	17	19	262
Gekas, G. PA	1.0949	-.1549	18	27	276
Rodino, P. NJ	1.0964	-.0899	19	59	283
Roberts, P. KS	1.0965	-.1330	20	35	251
Moore, H. LA	1.1667	-.1735	29	20	269
Long, G. LA	1.1818	-.2107	36	13	265
Pease, D. OH	1.2602	-.1835	70	16	165
Panetta, L. CA	1.3001	-.2854	89	2	204
Gingrich, N. GA	1.3189	-.1816	96	17	197
Kasich, J. OH	1.3189	-.2380	96	7	197
Fields, J. TX	1.4219	-.2500	147	5	144
Campbell, C. SC	1.4445	-.2183	155	10	130
Wolf, F. VA	1.4678	-.2127	166	12	115
Sikorski, G. MN	1.5168	-.2786	189	3	93
Erdreich, B. AL	1.5423	-.1861	201	14	85

An * indicates an individual among the most efficient using either TE or the residual from the TE regression.

Table 6. Least efficient incumbents as measured by technical efficiency scores^a and TE residuals

Name and state	Value		Rank		Vote
	TE	RESID	TE	RESID	
Albosta, D. MI	1.8570	-.0820	272	67	12
McCloskey, F. IN	1.8570	.0828	272	228	12
Clarke, J. NC	1.8570	.0149	272	168	12
Hansen, G. ID*	1.8570	.3104	272	284	12
Britt, C.R. NC	1.8570	.1007	272	240	12
Andrews, I. NC	1.8570	.0090	272	161	12
Long, C. MD	1.8957	-.0072	278	136	8
Patman, B. TX	1.8957	.0121	278	166	8
McNulty, J. AZ	1.8957	-.0050	278	139	8
Vandergriff, T. TX	1.8957	.0103	278	164	8
Crane, D. IL*	1.9361	.3570	282	286	7
Levitas, E. GA*	1.9782	.2242	283	277	5
Hightower, J. TX*	1.9782	.1847	283	272	5
Williams L. OH*	2.0222	.5704	285	288	3
Ratchford, W. CT*	2.0222	.3892	285	287	3
Patterson, J. CA	2.0683	.0555	287	205	1
Minish, J. NJ	2.0683	.1559	287	267	1
Clinger, W. PA	1.7844	.3216	263	285	24
Zschau, E. CA	1.4919	.1870	179	273	105
Ireland, A. FL	1.4919	.2164	179	276	105
Coats, D. IN	1.5168	.2539	189	282	93
Ford, W. MI	1.5426	.1789	201	270	85
Wilson, C. TX	1.5426	.2433	201	279	85
Derrick, B. SC	1.5689	.2371	209	278	73
Wortley, G. NY	1.6250	.2457	223	281	55
Levine, M. CA	1.6852	.1828	241	271	42
Gejdenson, S. CT	1.6852	.2062	241	275	42
Carney, W. NY	1.7170	.2884	248	283	35
Howard, J. NJ	1.7170	.1775	248	269	35
Hiler, J. IN	1.7501	.1991	255	274	27
Martinez, M. CA	1.7844	.2457	263	280	24

^aSix individuals are tied for 18th least efficient.

An * indicates the individual is among the least efficient using both TE and the residual from the TE regression.

trict, yet was able to win by campaigning on his legislative accomplishments and essentially ignoring the scandal. Studds' technical efficiency score was 1.6545 and his RESID was .024.

To test whether omission of a control for the scandals influenced the results, additional analysis was done in which a dummy variable for the presence of scandal was included.²⁰ The scandal variable was significant and positive. Moreover, the estimated residual for Representative Crane fell from .35695 to .16187, that of Representative Studds from .02480 to -.16187. These scandal controlled results still indict Dan Crane as an especially ineffective campaigner; his residual is only smaller than 20 other representatives, 6 of whom lost in the 1984 election. Gerry Studds appears as a very effective campaigner once controlling for the influence of the scandal; only 23 incumbents have lower residuals. For no other incumbent is the residual significantly altered. Consequently, it appears that the effect of scandal is evident in these residuals, but that fact alone is not sufficient to negate the argument that the residuals reflect candidate ability.

According to the *Almanac*, Don Albosta faced an insurmountable challenge in the reelection campaign of 1984. His well-financed challenger was the stepson of the chair of Dow Chemical, one leading employer in the district. Additionally, the challenger had been a campaign strategist for George Bush in 1980, showing experience in conducting a campaign. To top it off, he was good looking, and predisposed to the sort of flag-waving, celebration of America that characterized the Reagan presidential reelection campaign. Albosta, on the other hand, was a bit of a rabble rouser, who had investigated the Reagan campaign's obtaining of Carter campaign documents during the 1980 election. Moreover, he declared he believed the Reagan campaign had behaved illegally. Albosta went even further, suggesting that if what had been done was not illegal, then the law should be revised. The Congress failed to adopt his report, and even the Democratic party leadership distanced itself from Albosta's position. Consequently, he was not able to campaign as influential in Washington. Given all this, the *Almanac* remarks, "What's surprising, and shows Albosta's strength as a politician, is that the result was close." The estimated residual for Don Albosta is -.08202, the smallest of any incumbent candidate defeated in 1984 by a factor of more than 10.²¹

The focus of this section so far has been on losing incumbents. Consider now some winning incumbents. The most efficient incumbent, as measured by the residual, is Andrew Jacobs D-IN. His technical efficiency score and residual are, respectively, 1.0328 and -.3781. The *Almanac of American Politics 1986* describes Jacobs as an eccentric who disdains constituency services. Yet this lifelong Democrat represents Republican Indianapolis, though he has become more conservative overtime. He also takes relish in lampooning what

he perceives to be bad ideas whether the source is Republican or Democrat. Jacobs once refused to unseat a sitting committee chairman over whom he had seniority after losing his own chairmanship to a more senior colleague. His rationale was that experience in the committee chair was more valuable than musical chairmanships. The description of Andrew Jacobs in the *Almanac* suggests, to me at least, a principled individual who has the public interest at heart. The voters of Indianapolis, apparently, find those characteristics valuable, and the residual from the efficiency score reflects them.

Leon Panetta, D-CA, is the second most efficient campaigner according to the estimated residuals. The *Almanac of American Politics 1986* describes Representative Panetta as representative of his district. He is a former Republican, like many in his district, who is liberal on social issues. He had national prominence as an expert on the government's budget²² and had been very successful in supporting the agricultural interests of his district. The *Almanac* concluded that Panetta was "likely to continue winning by large margins." As in the cases described above, the residual from the efficiency score regression and the independent analysis agree that Leon Panetta is an effective campaigner and representative.

The upshot of these vignettes is that the residual from the regression explaining the technical efficiency score is remarkably consistent with the analysis in the *Almanac of American Politics 1986*. Consequently, concluding that these residuals reflect candidate quality either as a campaigner or a representative seems safe.

5. Conclusion

This paper has used data envelopment analysis to evaluate the efficiency with which congressional candidates use their spending to produce vote share. Efficiency scores were then examined using regression analysis to determine the influence of district and candidate characteristics on candidate efficiency. The results suggest that both district and candidate characteristics matter. Moreover, the results on district characteristics conform to intuition. For example, Democratic incumbents are more and Republican incumbents less efficient in districts that went more heavily in favor of Jimmy Carter. Nonwhite candidates are less efficient where the proportion of the population that is white is large.

The residuals from the efficiency score regressions could be interpreted in three ways, as candidate quality, as measures of omitted joint products such as pushing a legislative agenda or preparing for a run at higher office, or as an omitted variable such as political brand name capital. Wise use of limited campaign resources requires allocating funds to those candidates capable of

generating the biggest boost to vote share from a given amount of money. Consequently, political parties could use these residuals to determine which of their candidates, incumbents and/or challengers, are most likely to advance the party's cause most.

Two final points are in order. First, while the evidence here is strong, it has only been shown for one data set. It is desirable, therefore, that this style analysis be conducted on data from other election years, on both the House and the Senate, and even on state legislatures to check the robustness of the results. Additionally, if the residuals truly are measures of candidate ability as campaigners, as I have contended, then they should be positively correlated over time. This can be tested, and is the subject of ongoing research. Second, the data envelopment analysis of this paper is very crude and could and should be improved upon. A natural first step in this regard is to use more refined input measures. For example, if data on spending by type of input were available, it would be possible to determine the rate at which, say, use of television advertisements trades off with radio or print ads in the production of vote share. Such information would be very valuable in the discussion of campaign spending reform.

Notes

1. Green and Krasno (1988) is one, limited, exception. Coates (forthcoming) addresses this issue in greater detail.
2. Green and Krasno (1990) argue that comparison of incumbent and challenger marginal productivities is difficult in their work because they interact challenger quality with challenger spending, but do not interact incumbent spending with incumbent quality. They do not say so, but use of an index such as theirs for incumbent quality may be problematic. The variation across incumbents in the index will be small. See Green and Krasno (1988) for an explanation of index formation for challengers.
3. Charnes et al. (1994) and Seiford (1996) provide good introductions to the DEA literature.
4. In the DEA literature, a decision making unit, or observation, is referred to as a DMU.
5. Every maximization problem can be written alternatively as a minimization problem, and vice versa. It is often easier to solve the "dual" problem than the "primal."
6. The constraint that the λ 's sum to one allows the production surface to exhibit variable returns to scale. The constant ω is not restricted to zero in the variable returns case. One could impose constant returns on the model, which is equivalent to setting ω to zero.
7. The frontier extends horizontally from point C in this case because vote share, the output, cannot exceed 100%. In the more general case, this is not true. In fact, the surface may slope downward as some DMU may produce less output than the maximum observed output while using more inputs than the DMU with the highest output.
8. The efficiency scores utilized in this analysis are from an output maximization orientation. This means only that efficiency relates to the possible expansion of output using the current level of inputs. An alternative efficiency score is the ratio between the distance from the vertical axis to point D and the distance to point F. Such measures are referred to as input oriented because they focus on the least input for a given output.

9. Note that the TE are truncated in that $TE < 1$ are not observed. However, because some candidates not in the data may be more efficient than the units in the data set, it is possible that some TEs less than 1 exist. Hence, a tobit model is appropriate.

$$\begin{aligned} TE_i &= z_i\beta + \mu_i & \text{if } TE_i^* > 1 \\ TE_i &= 1 & \text{if } TE_i^* \leq 1 \end{aligned}$$

where TE_i^* is the unobservable “true” efficiency score. McCarty and Yaisawarng (1993) analyze this model for New Jersey school districts. Tobit analysis is not used here because there are so few efficient observations. Therefore, the limit variable is observed rarely in the data and Tobit results are essentially the same as those from ordinary least squares.

10. An alternative approach is to include these uncontrollable factors in the DEA. Banker and Morey (1986) and Golany and Roll (1993) demonstrate this approach and extensions to it.
11. Little of the published literature addresses or even acknowledges the role these characteristics play. A notable exception is Fenno (1978) which discusses personal characteristics of the legislator and how those characteristics match with the preferences of the constituency. However, neither the appropriate measurement of personal characteristics nor the means of rigorous empirical testing of the significance of the characteristics is indicated by Fenno’s analysis. The same is generally true for the theory of vote share production. Coates (forthcoming) is an exception. Consequently, estimation of the effects of these characteristics is ad hoc. The contribution of this paper is precisely that it offers an approach to this estimation that has not been used before and that has some intuitive appeal.
12. The DEA analysis was performed on two distinct models. The first model has only a single output, vote share of the candidate, and a single input, candidate spending. The second model adds an additional input variable, the size of the campaign war chest held by the candidate prior to the 1984 election. The results under either of these models are very nearly identical, so I focus on the former.
13. Under the alternative model in which war chest serves as an additional input in the DEA, there is a slight increase in the number of efficient observations. Four additional incumbents are efficient after the use of war chest as a DEA input: Cooper TN, Jacobs IN, Savage IL, Towns, NY.
14. Coates (1995) shows that personal characteristics explain a part of the “personal vote” of incumbent legislators; Coates (forthcoming) shows that the effects of spending vary with the characteristics of the legislator.
15. One might hypothesize that the manner in which district or personal characteristics variables influences campaign efficiency varies by party of the incumbent, as Ham Langford did when I presented this paper at Union College. I tested this by introducing interactions between the party dummy and the district and personal characteristics variables. The restriction that these additional variables have zero coefficients could not be rejected in the case of incumbents on whom the bulk of what follows focuses. Hence, these results are not reported.
16. If the dependent variable were vote share in the SUR regression, then the correlation in the errors would be (nearly) perfect since vote share across candidates must sum to one. Minor third party candidates and rounding error would suffice to break the perfect correlation. In the SUR reported in the text, the dependent variables are candidate efficiency scores, which need not be so highly correlated as vote shares. For instance, the comparison set of each may be quite different. The best observed practice against which the incumbent and challenger are compared need not be correlated.

17. A test of the difference in the means is meaningless since by construction the mean for each group is zero. It is, after all, the average of the OLS residuals.
18. Using the Lagrange Multiplier test described in Greene (1990), the test statistic is 108.3 and the critical value of the chi squared distribution with one degree of freedom is 3.84.
19. The complete list is available on request.
20. The variable took the value 1 for Studds and Crane and 0 for everyone else. This approach is similar to that of Kevin Grier (1989).
21. The next smallest residual was $-.0072$.
22. This expertise earned him an appointment in the Clinton Administration as Director of the Office of Management and Budget.

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